



D Y PATIL INTERNATIONAL UNIVERSITY

Sector 29, Pradhikaran, Akurdi, Pune - Maharashtra, INDIA 411044

(Establishment by Maharashtra Act No. LXIII of 2017)

School of Computer Science, Engineering & Applications

Course Name	Fundamentals of Generative Artificial Intelligence
Course Code	AIM3002P
Credits	1
Prerequisites	Basics of ML, DL and AI
L-T-P-C	0-0-2-1

COURSE OBJECTIVES

1. To explore generative AI tools and apply prompt engineering techniques for text generation.
2. To implement and analyze generative models including VAEs, GANs, and diffusion models.
3. To apply fine-tuning techniques and develop applications using Large Language Models.
4. To deploy and evaluate generative AI applications using appropriate tools and evaluation metrics.

LABORATORY

Sr. No.	Content
1	Introduction to GenAI Tools: Install and explore Hugging Face Transformers, generate text with GPT-2, visualize discriminative vs. generative outputs.
2	Basic Prompt Engineering: Implement basic and instruction-based prompting with DistilGPT-2.
3	Advanced Prompt Techniques: Apply Chain-of-Thought (CoT) and role-based prompting, evaluate manually.
4	Build a Simple VAE: Load a pre-trained VAE for MNIST reconstruction, visualize latent space.
5	Simple GAN Exploration: Load a pre-trained DCGAN, generate and visualize samples.
6	Image Generation with Diffusion: Use pre-trained Stable Diffusion for text-to-image generation.
7	LLM Fine-Tuning Basics: Lightly fine-tune DistilBERT with LoRA on a small dataset.
8	Build a Basic Chatbot with LangChain: Integrate LLaMA-2 with LangChain for a Q and A chatbot.
9	Deployment and Evaluation: Deploy Stable Diffusion via Gradio with a safety filter, evaluate with FID.

COURSE OUTCOMES

At the end of the course the student will be able to:

CO1	Explore generative AI tools and generate content using pre-trained generative models.
CO2	Apply basic and advanced prompt engineering techniques for effective interaction with generative models.
CO3	Implement and analyze generative models including VAEs, GANs, and diffusion models.
CO4	Apply fine-tuning techniques and develop applications using Large Language Models.
CO5	Deploy and evaluate generative AI applications using appropriate tools, safety mechanisms, and evaluation metrics.

CO-PO Mapping:		3: Substantial					2: Moderate					1: Slight		
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	2	3	-	-	-	-	-	-	2	3	2
CO2	2	3	3	2	3	-	-	-	-	-	-	2	3	2
CO3	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO4	3	3	3	3	3	-	-	-	-	-	2	3	3	3
CO5	3	3	3	3	3	2	-	-	2	2	2	3	3	3

TEXT BOOKS:

1. David Foster, *Generative Deep Learning*, 2nd Edition, O'Reilly Media, Inc., ISBN: 9781098134181, 2023.
2. James Phoenix, Mike Taylor, *Prompt Engineering for Generative AI*, O'Reilly Media, Inc. ISBN: 9781098153434, 2024
3. Dr. Priyanka Singh and Hariom Singh, *Generative AI Essentials: Unlocking Creativity and Innovation with Generative AI*, BPB Publications

REFERENCE BOOKS:

1. Russell and Norvig., *Artificial Intelligence: A modern Approach*, Printice Hall, 2010.
2. Dan W. Patterson, *Introduction To Artificial Intelligence and Expert Systems*, PHI Publications.
3. Denis Rothman, *Transformers for Natural Language Processing*, Packt Publishing (2021).

EVALUATION SCHEME

Continuous Assessment			End-term Practical	Total
Lab Manual	Lab Assessment	Internal Viva-voce		
10	10	30	50	100